



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of methodologies:**

SpaceX's Falcon 9 First stage landing success prediction is complex task that involves analysis of various factors. The methodologies used to predict the landing success includes;

- ❖ **Data Collection:** Data was collected from SpaceX's API and web scraping, Information from rocket launches, payloads, launchpads, payloads and cores were collected
- ❖ **Data Wrangling:** Preprocessing, transformation and one-hot encoding was carried out on the data collected to prepare it for machine learning models.
- ❖ **Exploratory Data Analysis(EDA):** Various Visualization using SQL and techniques such as scatter plots, bar charts, line charts, maps, and interactive dashboards to identify patterns and correlation in data.
- ❖ **Predictive Analysis:** Classification machine learning models were built to predict the landing success and the models were evaluated using metrics such as accuracy, precision and recall.
- ❖ **Feature Engineering:** Launch site, payload mass and orbit type were engineered to improve the accuracy of the models
- ❖ **Model Selection:** The best model was selected based on its performance on the test data and Decision tree model achieved the highest accuracy of 0.889.
- ❖ **Hyperparameter Tuning:** Gridsearch was used to perform hyperparameter tuning to optimize models performance.

Executive Summary

Summary of all results

- **Accuracy Rates:** In the study, the highest accuracy of 89% was achieved using a Decision Tree model. In other studies, LogisticRegression Classifier, K-nearest neighbors, Support Vector Machines achieved accuracy of 83%.
- Key Factors associated with the success rate of Falcon 9 rocket launches includes; Launch site, Payload mass, Orbit type and Year of launch.

Introduction

Globally, Space technology is fast becoming a go to when it comes to global communication, navigation and exploration and therefore, the success of launch missions has become paramount. SpaceX's Falcon 9 rocket has appeared as a dependable workhorse in the space industry owing to its major advantage of low cost when compared to other providers in the industry.

Though, SpaceX's Falcon 9 has achieved numerous successful launches, but predicting the outcome of launches remains a daunting task. The aim of this project is to develop a predictive model that can accurately forecast the landing of the first stage and determine the cost of launch based on historical launch data and relevant factors.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - The data was sourced from SpaceX API by making call request to SpaceX API
- Perform data wrangling
 - Columns with missing values were identified and replaced with the mean value of the column
- Perform exploratory data analysis (EDA) using visualization and SQL

Various exploratory data analysis were performed on dataset to identify trends

- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

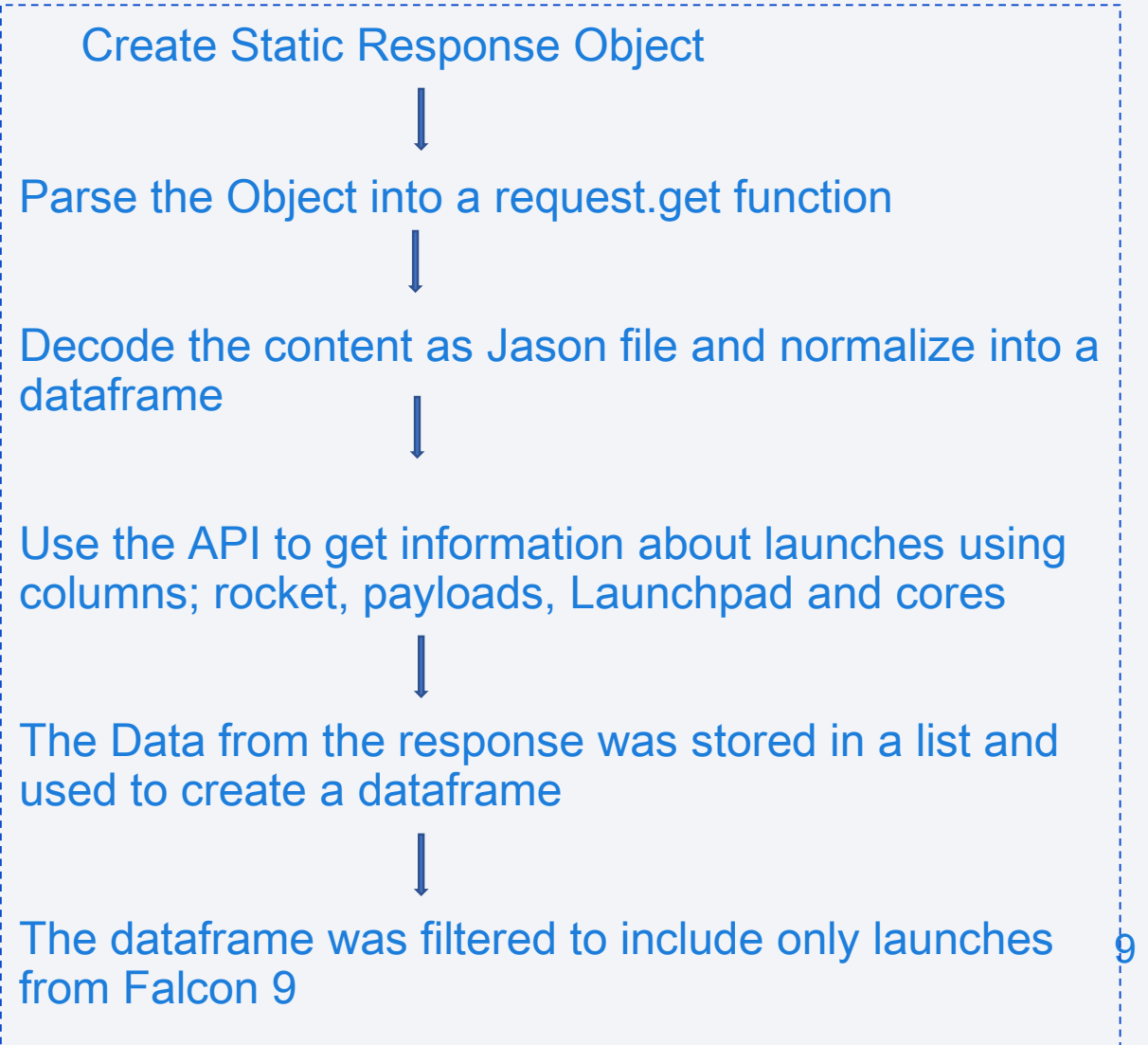
Data Collection

Data was collected from the SpaceX API using the following steps;

- ✓ API Request: Request was sent to SpaceX API using get request function.
- ✓ The response was decoded as Jason file normalized into a data frame.
- ✓ The API was used again to get specific information about the launches using columns rockets, payloads, Launchpad and cores.
- ✓ The data from the above request was stored in a list and used to create a data frame
- ✓ The data frame created was filtered to include only the Falcon 9 launches.

Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
- Github URL for data collection notebook
(<https://github.com/Esomsamuel/estrepo2/compare/main...Esomsamuel-Cousera-CapstoneProject-Data-Collection?expand=1>)



Data Collection - Scraping

- GitHub URL of the completed web scraping notebook
(<https://github.com/Esomsmuel/testrepo2/upload/main>)

Create a static url object using url for list of falcon 9 and falcon 9 heavy launches wikipedia



Pass the static url object into a request.get function



Create a BeautifulSoup object for the HTML response from the get request of the above.



The column/variable names were extracted from the HTML table header



Create a dataframe by parsing the launch HTML table

Data Wrangling

- The data were processed by identifying the row with missing values

Data wrangling flowchart.

The mean of the PayloadMass was calculated



The mean was used to replace the missing numbers using replace function.

- GitHub URL of your completed data wrangling related notebooks(<https://github.com/Esomsamuel/testrepo2/upload/main>)

EDA with Data Visualization

- The following charts were used in this project;

- scatter plots

- Barcharts

- Line charts.

GitHub URL of EDA with data visualization notebook
(<https://github.com/Esomsamuel/testrepo2/tree/Esomsamuel-Couser-CapstoneProject-Data-Collection>)

EDA with SQL

- The following SQL queries you performed

- Query to displace distinct launch sites
- Query to display 5 records where the launch site begins with 'CCA'
- Query to display the total payload mass carried by boosters launched by NAS (CRS)
- Query to display the average payload mass carried by booster version f9 v1.1
- Query to list the date when the first successful landing outcome in ground pad was achieved.
- Query to list the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- Query to list the total successful and failure mission outcomes
- Query to display the month names, failure landing outcomes in drone ship ,booster versions, launch site for the months in year 2015
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

GitHub URL of completed EDA with SQL notebook

<https://github.com/Esomsamuel/testrepo2/tree/Esomsamuel-CouserCapstoneProject-Data-Collection>

Build an Interactive Map with Folium

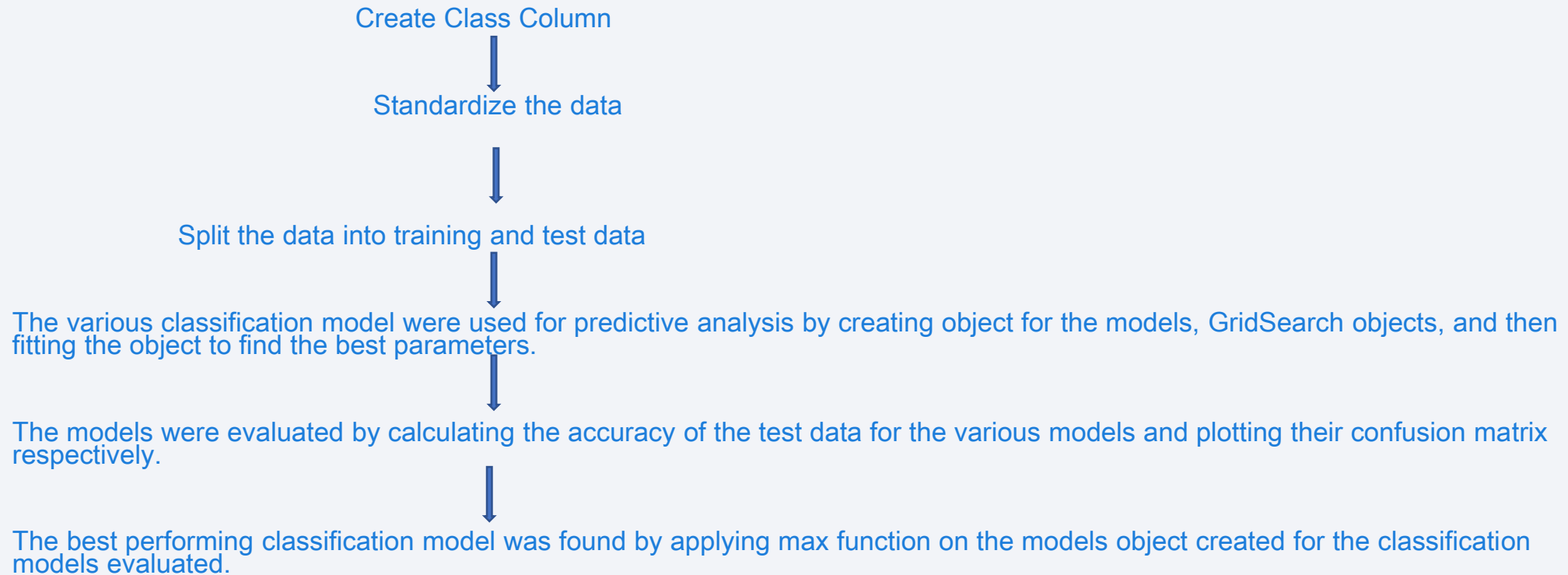
- Folium markers, Marker colors, circles and Polyline were created and added to the folium map.
- ❖ Blue circle at NASA Johnson space center's coordinate with pop-up label showing its center was created.
- ❖ **Marker color:** This was created indicate the success and failure of each launches for each launch sites
- ❖ **Polyline:** This was created and added to the folium map to determine the proximities between a launch site and a selected coastline.
- GitHub URL of completed interactive map with Folium map (<https://github.com/Esomsamuel/testrepo2/tree/Esomsamuel-Couser-CapstoneProject-Data-Collection>)

Predictive Analysis (Classification)

- Class column was created, the data standardized and then split into training and test data. Objects were created for Logistic regression, Support vector machine, Decision tree classifier and K-nearest neighbour followed by GridSearch object with $cv = 10$. These objects were then fit to find the best parameters from the dictionary 'parameters'. The various models were evaluated by calculating the accuracy of the test data and plotting the confusion matrix for each of model, improved, and the best performing classification model was found by applying the max function
- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose

Predictive Analysis (Classification)

- Flowchart of Model Development



- The GitHub URL of predictive analysis lab (<https://github.com/Esomsamuel/testrepo2/tree/Esomsamuel-Couser-CapstoneProject-Data-Collection>)

Results

- Exploratory data analysis results

Names of the unique launch site the space mission - ('CCAFS LC-40'), ('VAFB SLC-4E'), ('KSC LC-39A'), ('CCAFS SLC-40'),

Five records where launch site begins with CCA –

('2010-06-04', '18:45:00', 'F9 v1.0 B0003', 'CCAFS LC-40', 'Dragon Spacecraft Qualification Unit', 0, 'LEO', 'SpaceX', 'Success', 'Failure (parachute)')

('2010-12-08', '15:43:00', 'F9 v1.0 B0004', 'CCAFS LC-40', 'Dragon demo flight C1, two CubeSats, barrel of Brouere cheese', 0, 'LEO (ISS)', 'NASA (COTS) NR O', 'Success', 'Failure (parachute)')

('2012-05-22', '7:44:00', 'F9 v1.0 B0005', 'CCAFS LC-40', 'Dragon demo flight C2', 525, 'LEO (ISS)', 'NASA (COTS)', 'Success', 'No attempt')

('2012-10-08', '0:35:00', 'F9 v1.0 B0006', 'CCAFS LC-40', 'SpaceX CRS-1', 500, 'LEO (ISS)', 'NASA (CRS)', 'Success', 'No attempt')

('2013-03-01', '15:10:00', 'F9 v1.0 B0007', 'CCAFS LC-40', 'SpaceX CRS-2', 677, 'LEO (ISS)', 'NASA (CRS)', 'Success', 'No attempt')

Total payload mass carried by boosters launched by NASA (CRS) – (45596)

Date when the successful landing outcome for ground pad was achieved – (2015-12-22)

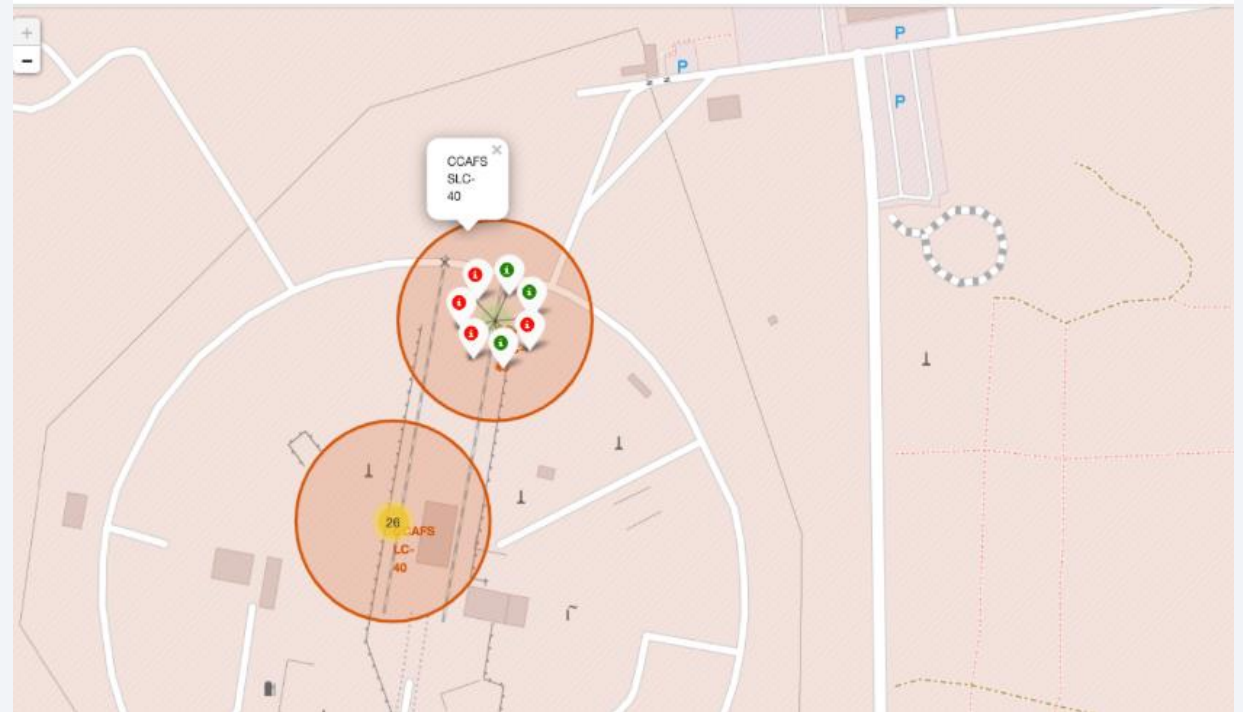
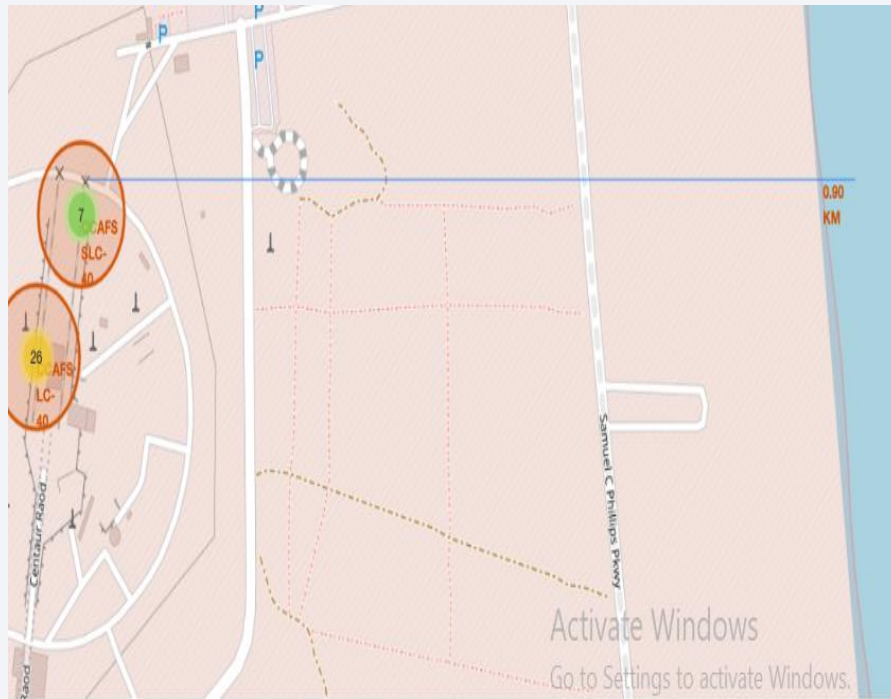
Names of boosters which have success in drone ship with payload mass greater than 4000 but less than 6000 – [('F9 FT B1022') ('F9 FT B1026'), ('F9 FT B1021.2'), ('F9 FT B1031.2')]

Total number of success and failures mission outcomes – (98, 0) respectively

List of booster versions that have carried maximum payload mass - [('F9 B5 B1048.4'), ('F9 B5 B1049.4'), ('F9 B5 B1051.3'), ('F9 B5 B1056.4'), ('F9 B5 B1048.5'), ('F9 B5 B1051.4'), ('F9 B5 B1049.5'), ('F9 B5 B1060.2 '), ('F9 B5 B1058.3 '), ('F9 B5 B1051.6'), ('F9 B5 B1060.3'), ('F9 B5 B1049.7 '),)]

Results

- Interactive analytics demo in screenshots



Results

Predictive analysis results

Accuracy test score for logreg model = 0.83333333333333333334

Accuracy test score for support vector machine model = 0.83333333333333333334

Accuracy test score for decision tree classifier model = 0.72222222222222222222

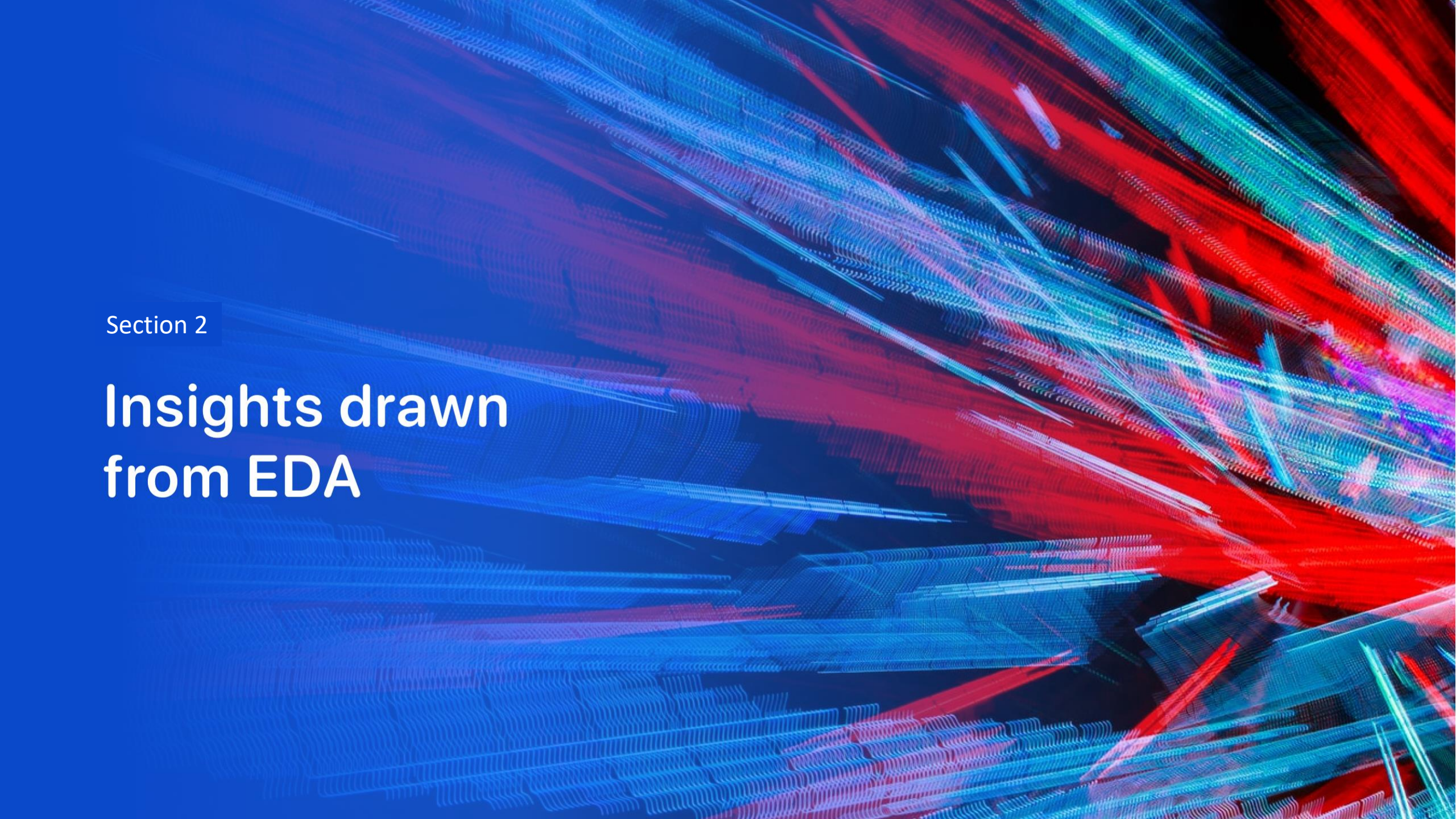
Accuracy test score for K nearest neighbor model = 0.83333333333333333334

Best performing model;

Best Model: Decision Tree

Best Score: 0.8892857142857142

Best Parameters: {'criterion': 'entropy', 'max_depth': 8, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 10, 'splitter': 'rando

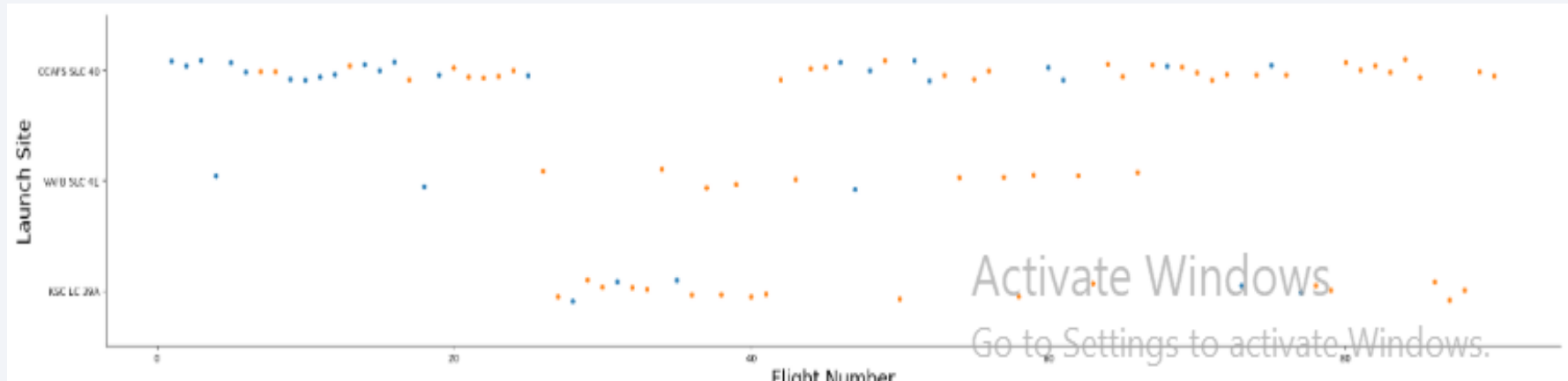
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

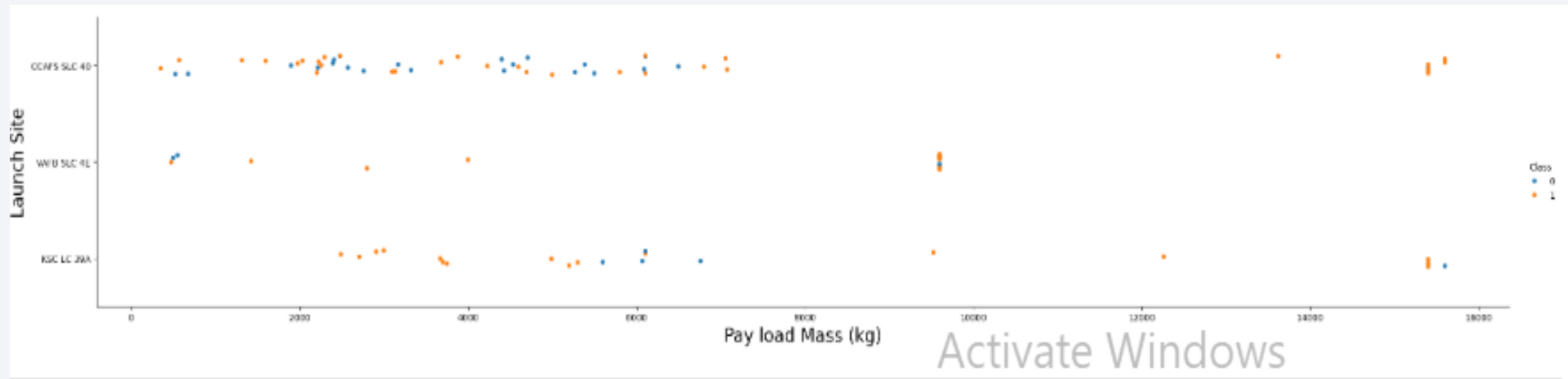
- Scatter plot of Flight Number vs. Launch Site



- The scatter plot shows that there is no linear relationship between Flight Number and Launch Site

Payload vs. Launch Site

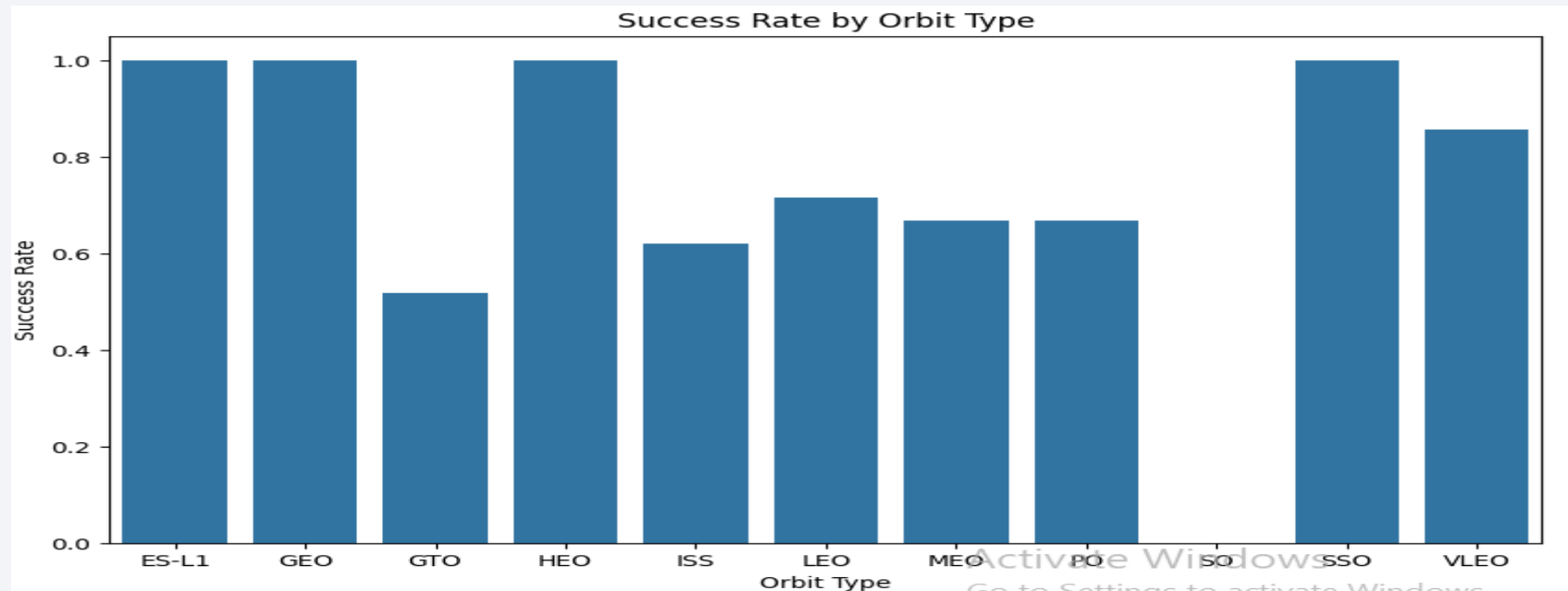
- Scatter plot of Payload vs. Launch Site



- From the scatter plot it can be observed that for VAFB-SLC there are no rockets launched for heavy payload mass (greater than 10000)

Success Rate vs. Orbit Type

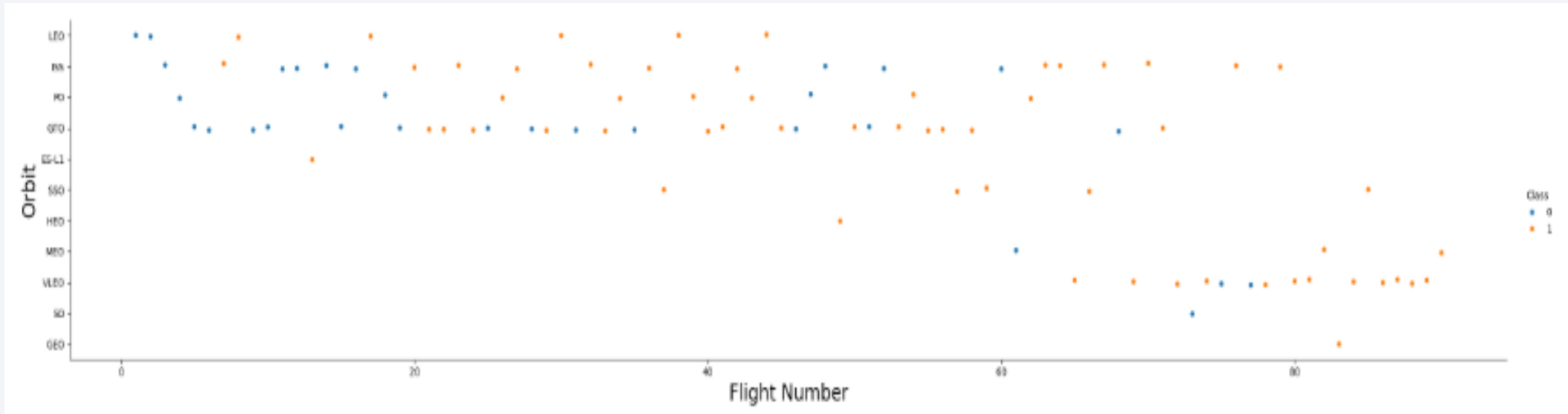
- Bar chart for the success rate of each orbit type



- The bar chart plot shows that the orbits ES-L1, GEO, HEO and SSO has the highest success rate.

Flight Number vs. Orbit Type

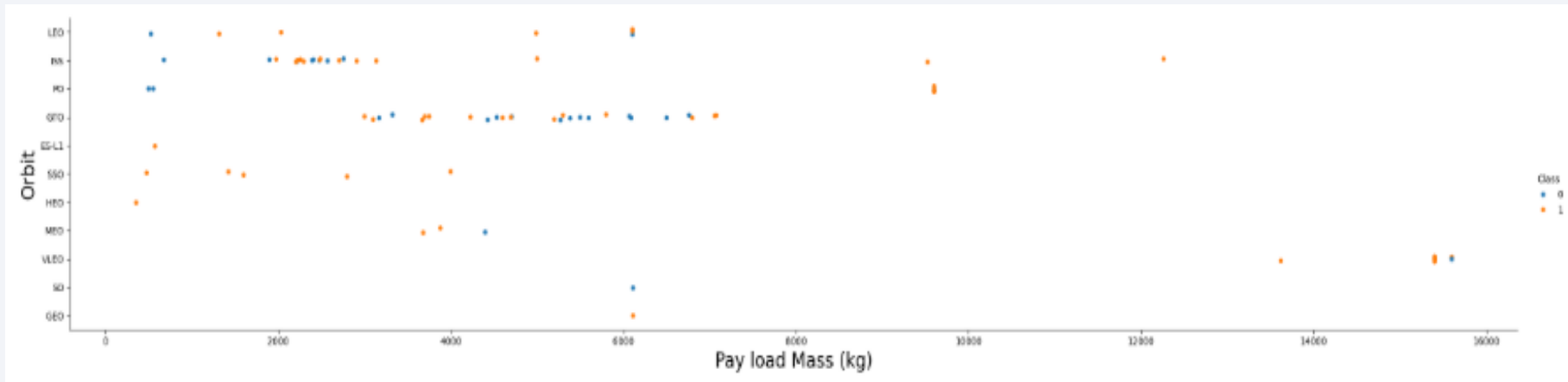
- Scatter point of Flight number vs. Orbit type



- It can be observed from the plot that in the LEO orbit, success seems to be related to the number of flight, whereas in the GTO orbit, there appears to be no relationship between flight number and success.

Payload vs. Orbit Type

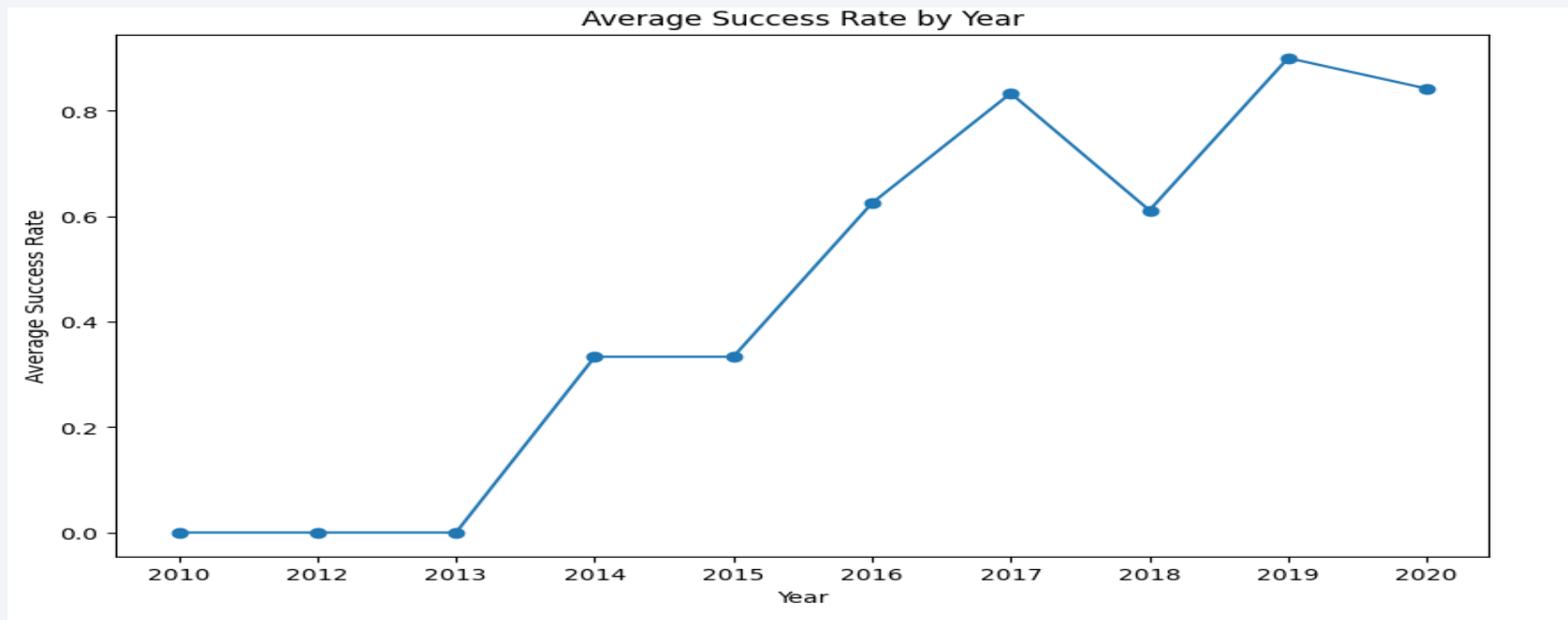
- Scatter point of payload vs. orbit type



- Successful landing for heavy payload are more for Polar, LEO and ISS. However, with GTO, its difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend

- Line chart of yearly average success rate



- From the line chart it can be seen that there has been an increase in success rate from 2013 till 2020

All Launch Site Names

The names of the unique launch sites includes;

- CCAFS LC-40
- VAFB SLC-4E
- KSC LC-39A
- CCAFS SLC-40

CCAFS LC-40 and CCAFS SLC-40 are used to launch Falcon 9 into orbit, VAFB SLC-4E used to launch satellites into polar or sun-synchronous orbits using Falcon 9, KSC LC-39A support starship and crewed mission.

Launch Site Names Begin with 'CCA'

5 records where launch sites begin with `CCA`

- ('2010-06-04', '18:45:00', 'F9 v1.0 B0003', 'CCAFS LC-40', 'Dragon Spacecraft Qualification Unit', 0, 'LEO', 'SpaceX', 'Success', 'Failure (parachute)')
- ('2010-12-08', '15:43:00', 'F9 v1.0 B0004', 'CCAFS LC-40', 'Dragon demo flight C1, two CubeSats, barrel of Brouere cheese', 0, 'LEO (ISS)', 'NASA (COTS) NRO', 'Success', 'Failure (parachute)')
- ('2012-05-22', '7:44:00', 'F9 v1.0 B0005', 'CCAFS LC-40', 'Dragon demo flight C2', 525, 'LEO (ISS)', 'NASA (COTS)', 'Success', 'No attempt')
- ('2012-10-08', '0:35:00', 'F9 v1.0 B0006', 'CCAFS LC-40', 'SpaceX CRS-1', 500, 'LEO (ISS)', 'NASA (CRS)', 'Success', 'No attempt')
- ('2013-03-01', '15:10:00', 'F9 v1.0 B0007', 'CCAFS LC-40', 'SpaceX CRS-2', 677, 'LEO (ISS)', 'NASA (CRS)', 'Success', 'No attempt')
- The query result selected launch data from spacex_launches table filtering only launch sites starting with CCA and limiting the number to only 5

Total Payload Mass

The total payload carried by boosters from NASA

- 45596
- The query selects all launch data with payload mass for NASA and uses the SUM aggregate function to add up the payload masses

Average Payload Mass by F9 v1.1

The average payload mass carried by booster version F9 v1.1 is 2928

- The query selects the average PAYLOAD_MASS_KG using the function AVG.

First Successful Ground Landing Date

The date of the first successful landing outcome on ground pad is
2015-12-22

- From the query result, it can be seen that the first successful landing on ground pad from the spacex table occurred in 20-12-2022.

Successful Drone Ship Landing with Payload between 4000 and 6000

- Names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000 includes;
 - F9 FT B1022
 - F9 FT B1026
 - F9 FT B1021.2
 - F9 FT B1031.2
- A total of four (04) boosters with payload mass greater than 4000 but less than 6000 successfully landed on drone ship according to the SQL query.

Total Number of Successful and Failure Mission Outcomes

There were a total of 98 successful mission outcomes and Zero i.e no failure mission outcome according to the result returned by SQL query on SPACEXTABLE.

- The result from the query indicates that all the mission outcomes were successful.

Boosters Carried Maximum Payload

- The booster which have carried the maximum payload mass includes;
 - F9 B5 B1048.4
 - F9 B5 B1049.4
 - F9 B5 B1051.3
 - F9 B5 B1056.4
 - F9 B5 B1048.5
 - F9 B5 B1051.4
 - F9 B5 B1049.5
 - F9 B5 B1060.2
 - F9 B5 B1058.3
 - F9 B5 B1051.6
 - F9 B5 B1060.3
 - F9 B5 B1049.7
- Twelve boosters according to the query result carried the maximum payload mass

2015 Launch Records

- The failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015 includes

S/N	LANDING OUTCOMES	BOOSTER VERSIONS	LAUNCH SITE
1.	'01', 'Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
2.	'04', 'Failure (drone ship)'	F9 v1.1 B1015	CCAFS LC-40

- The failure landing outcome in drone ship is the first landing at booster version F9 v1.1 B1012 at launch site CCAFS LC-40 and another failure landing outcome of drone ship is at the fourth landing at booster version F9 v1.1 B1015 and launch site CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The Landing outcomes count (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order;

'Success (drone ship)', 5

↓
'Failure (drone ship)', 5

↓
'Success (ground pad)', 3

↓
'Controlled (ocean)', 3

↓
'Uncontrolled (ocean)', 2

↓
'Failure (parachute)', 2

↓
Precluded (drone ship)', 1

- The landing outcomes count ranged from 5 success drone ship to 1 precluded 1 drone ship

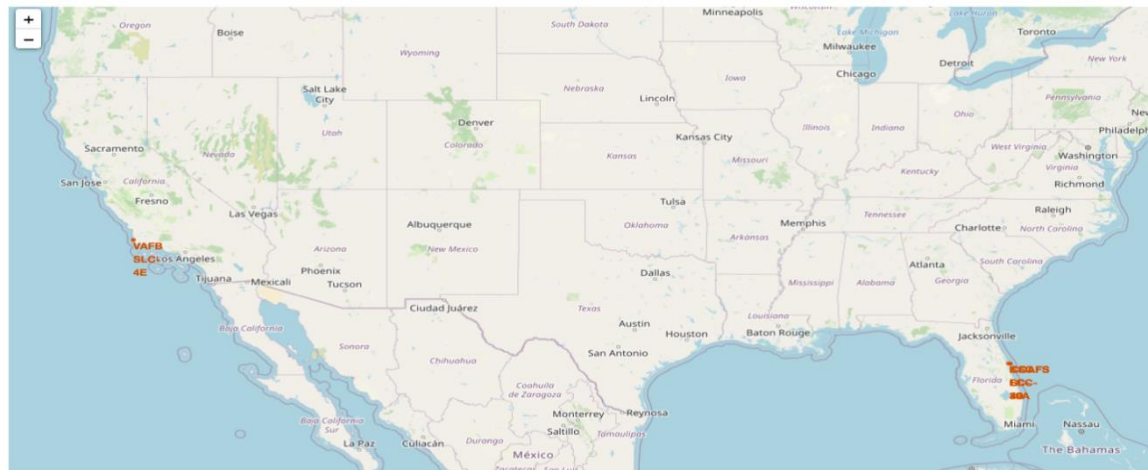
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

LAUCH SITES ON THE MAP

- Map showing all launch sites



- According to the folium map, all launch site are close to equator to minimize fuel consumption by using earth's $\sim 30\text{km/s}$ eastward spin to help spaceship get into orbit. Also, Most of the launch site are close to the coast so crew can have option to abort launch and attempt water landing and minimize people and property risk from falling debris

LAUNCH OUTCOME ON MAP

- Map showing launch outcomes



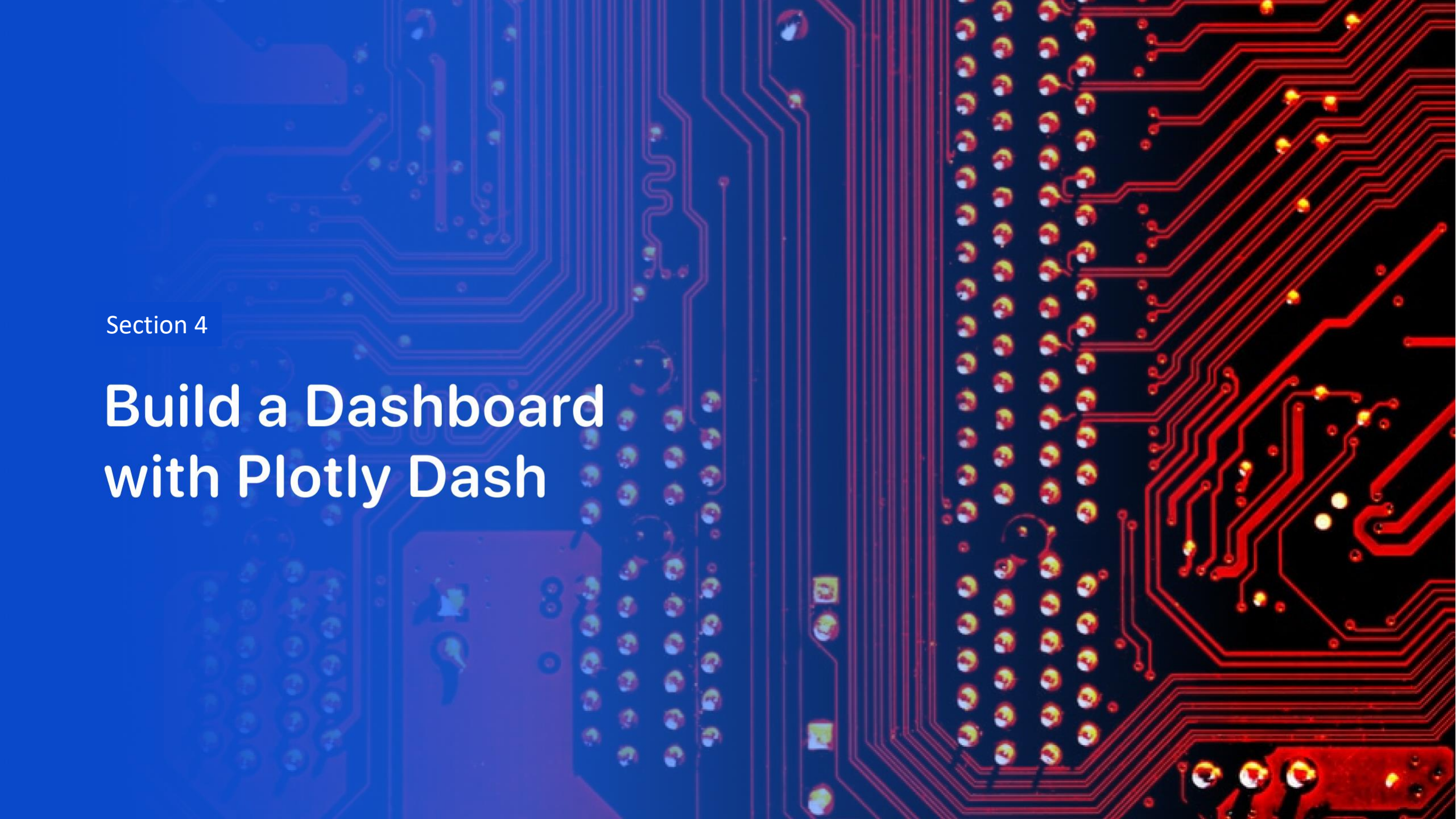
- Explore the folium map and make a proper screenshot to show the color-labeled launch outcomes on the map
- Explain the important elements and findings on the screenshot

LAUNCH SITE TO ITS PROXIMITIES

- Map showing launch site to its proximities



- Launch site are in close proximity to highways for easy transport required for people and property. Close to railway for heavy cargo transport and close to coastline for safety reasons.

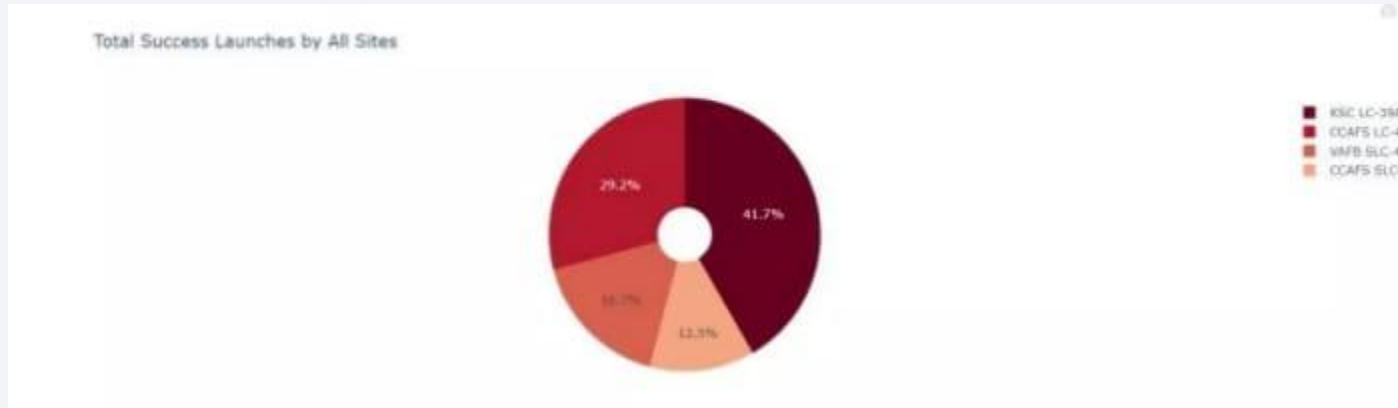


Section 4

Build a Dashboard with Plotly Dash

LAUNCH SUCCESS COUNT

- Launch Success Count for all sites



KSC LC-39A has the highest success score with 41.7%

CCAFS LC-40 is the second with 29.2%

VAFB SLC-4E and CCAFS SLC-40 launch sites has 16.7% and 12.5% respectively.

LAUNCH SITE WITH THE HIGHEST SCORE

Launch site with the highest score.

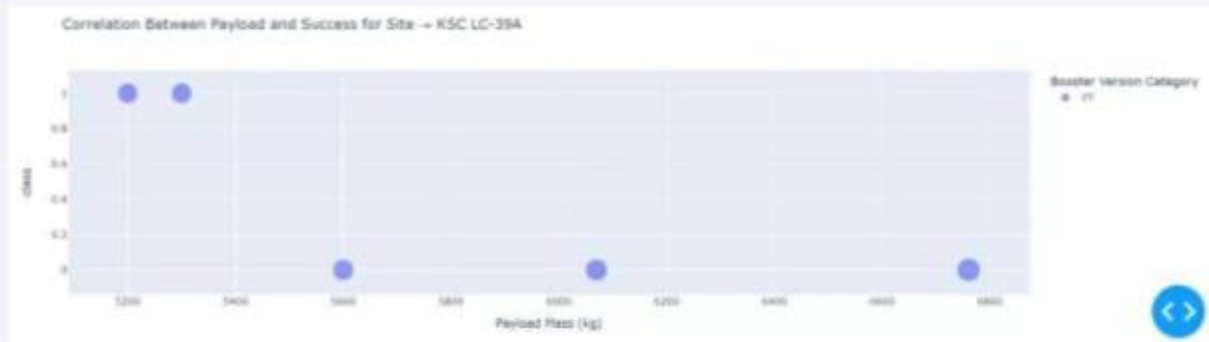


KSC LC-39A has the highest score with 76.9% with payload range of 2000kg – 10000kg and FT booster version has the highest score

PAYLOAD VS. LAUNCH OUTCOME



Payload 0 kg – 5000kg
(first half)



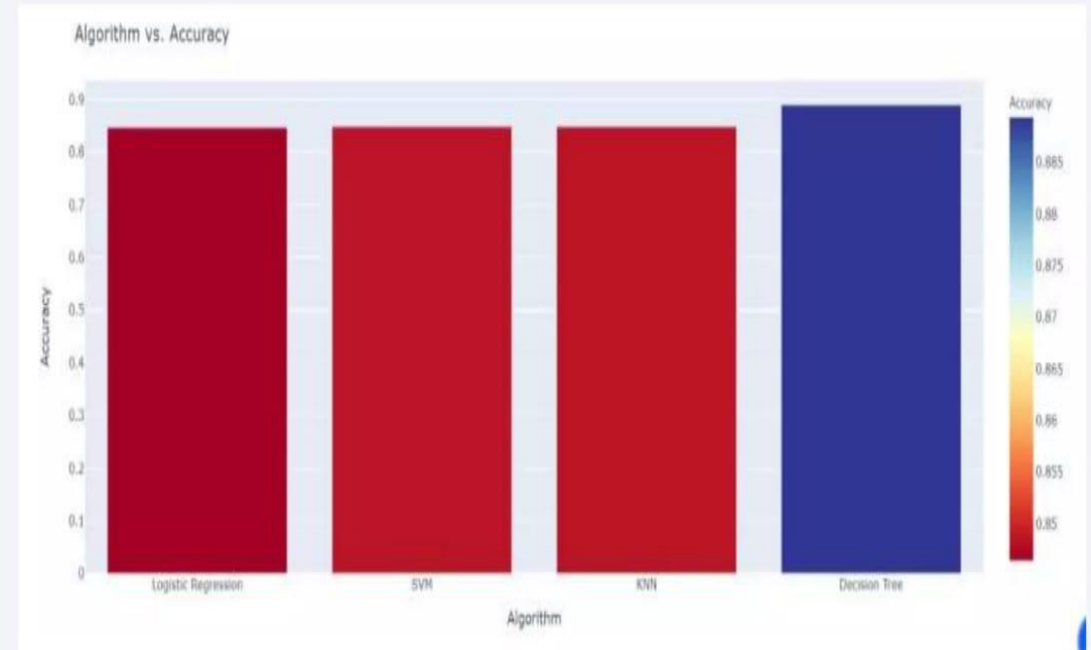
Payload 6000k - 10000kg
(Second half)

Section 5

Predictive Analysis (Classification)

Classification Accuracy

- The image on the right is a bar chart comparing the various model accuracy.
- The Decision Tree Classifier produced the best score of approximately 0.89 of all the models deployed to predict the outcomes.

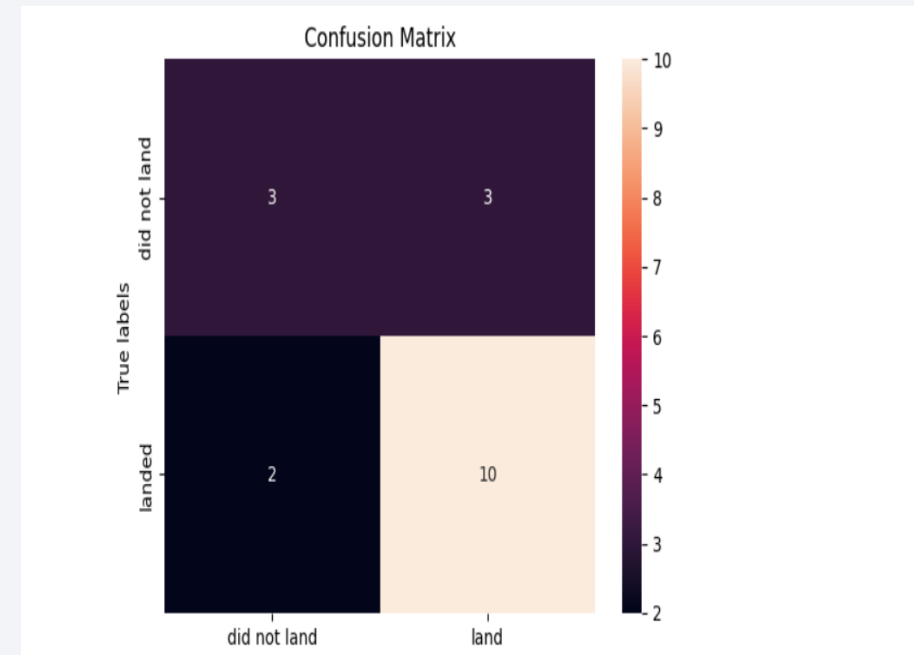


Confusion Matrix

- The confusion matrix of the Decision Tree Classifier the best performing model.

From the predicted value, the model;

- Correctly predicted 3 positive class
- Correctly predicted 10 negative class
- Incorrectly predicted 3 positive class when the actual outcome was negative.
- Incorrectly predicted 2 negative class when the actual outcome was positive.



Conclusions

- This project predicted if the first stage landing of a given Falcon 9 launch will land in order to determine the cost of the launch.
- It was determined from the project that each feature of the Falcon 9 such as its payload mass and orbit type may affect the mission outcome.
- Several machine learning models were deployed to predict the outcome of Falcon 9 launch
- The Decision Tree classifier produced the Best Score of 0.889285714287142 indicating it to be the best performing model among the 4 machine model deployed to predict the outcome of the launch.

Appendix

My codes snippets can be found on all my uploaded notebooks on GitHub

GitHub repo: <https://github.com/Esomsamuel/testrepo2/upload/main>

Thank you!

